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Winsyahputra Ritonga
State University of Medan
Ahmad Andi Solahuddin
State University of Medan
Budiman Nasution
budimannasution@unimed.ac.id
State University of Medan
Habibi Azka Nasution
State University of Medan

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A Hybrid Physics-Informed Deep Learning Framework for Predictive Diagnostics of Concrete Fracture

Winsyahputra Ritonga[1*], Ahmad Andi Solahuddin[2†], Budiman Nasution[1*†], Habibi Azka Nasution[1†]

[1*]Department of Physics, Universitas Negeri Medan, Willem Iskandar Street, Pasar V Medan Estate, Medan, 20221, North Sumatera, Indonesia.

[2]Department of Building Engineering Education, Universitas Negeri Medan, Willem Iskandar Street, Pasar V Medan Estate, Medan, 20221, North Sumatera, Indonesia.

*Corresponding author(s). E-mail(s): winsyahputra@unimed.ac.id; budimannasution@unimed.ac.id;
Contributing authors: ahmad.andi@unimed.a.id; habibiazka@unimed.ac.id;
†These authors contributed equally to this work.

Abstract

This study presents a hybrid physics-informed deep learning framework for quantitative diagnostics of concrete cracks. Integrating finite element simulations with convolutional neural networks (CNNs), the model predicts key fracture mechanics parameters-Stress Intensity Factor ($\boldsymbol{K_I}$) and Energy Release Rate (G-directly from stress-field images. Crucially, a custom loss function enforces the fundamental physical constraint $\boldsymbol{G = (\frac{K_I^2)}{E}}$ during training, ensuring predictions are not only accurate but also physically consistent. Despite the constrained dataset ($\boldsymbol{n = 50}$), the model demonstrated stable convergence and successfully learned the underlying fracture patterns. Crucially, the enforced physical constraint $\boldsymbol{G = \frac{(K_I^2)}{E}}$ proved highly effective as a powerful regularizer, which is evident in the controlled gap between the training and validation loss. The final test set evaluation yielded a Mean Absolute Error (MAE) of 3.7691. While this error is

high due to data scarcity, the results successfully validate the framework's feasibility and physical fidelity, providing a trustworthy, first-principles-based tool for structural health monitoring beyond purely data-driven methods.

Keywords: concrete cracks, physics-informed AI, FEM, fracture mechanics, machine learning

1 Introduction

Concrete is the most widely used material in modern infrastructure, yet its susceptibility to cracking undermines durability and safety. Conventional crack detection methods rely on manual inspection or image-based AI, but these approaches often neglect the governing physics of fracture mechanics [1][2][3][4][5][6][7][8]. Purely physics-based approaches, such as LEFM and finite element simulations, provide deep insights into crack behaviour but lack scalability for real-time diagnostics. On the other hand, AI methods excel in processing large datasets but often act as black boxes without physical interpretability.

The widespread use of concrete stems from its abundance, low cost, high compressive strength, and adaptability to diverse structural forms, making it the backbone of bridges, buildings, dams, and pavements. However, concrete is inherently brittle and prone to cracking due to several factors: shrinkage during curing, thermal stresses, cyclic mechanical loading, chemical degradation such as alkali?silica reaction, and environmental effects like freeze-thaw cycles. Cracks not only compromise the structural integrity but also accelerate the ingress of water, chlorides, and other aggressive agents, ultimately reducing service life and increasing maintenance costs [9][10][11][12].

Consequently, early crack detection and monitoring are critical to ensure safety, extend structural durability, and reduce repair costs. Previous research has addressed this challenge from different angles. Manual inspection and image-based computer vision approaches provide practical solutions but are limited by subjectivity and poor generalizability [13][14][15]. Physics-based models, including linear elastic fracture mechanics (LEFM) and finite element method (FEM) simulations [16][17][18][19][20], have advanced understanding of crack initiation and propagation, though their computational intensity limits real-time applications. More recently, machine learning (ML) and deep learning methods have shown remarkable promise in automating crack detection from large datasets [21][22][23][24], yet they often lack physical grounding, which restricts their interpretability and reliability under unseen conditions. This motivates the development of a hybrid framework that integrates fracture mechanics with AI, combining the interpretability of physics with the adaptability of data-driven learning.

A research gap exists in combining physics-based fracture mechanics with machine learning to develop interpretable, accurate, and scalable crack diagnostics. This study proposes a hybrid framework that bridges the gap by embedding physical equations within AI models and using finite element simulations to generate physics-consistent datasets.

2 Theoretical Background

2.1 Linear Elastic Fracture Mechanics (LEFM)

Linear Elastic Fracture Mechanics (LEFM) provides the fundamental theoretical basis for analyzing crack behavior in brittle and quasi-brittle materials like concrete. Its central premise is that the stress field near a sharp crack tip is uniquely characterized by a single parameter, the Stress Intensity Factor (SIF) [25][26][27]. For Mode-I (opening mode) loading, the SIF is given by:

$$K_I = \sigma\sqrt{\pi a} \cdot Y(\frac{a}{W}). \tag{1}$$

Where σ is the applied remote stress, a is the crack length, W is the specimen width, and Y is a dimensionless geometry correction factor that accounts for finite boundaries and loading configurations. Crack propagation initiates when the driving force K_I equals or exceeds the material's fracture toughness K_IC, a critical intrinsic property $K_I \geq K_{IC}$.

An equivalent energy-based approach describes crack extension through the Energy Release Rate G, defined as the rate of change in potential energy with respect to crack area. For linear elastic materials, G and K_I are fundamentally interrelated:

$$G = \frac{K_I^2}{E'}. \tag{2}$$

Here, E' is the effective elastic modulus, defined as E for plane stress conditions and $\frac{E}{1-v^2}$ for plane strain, with v as Poisson?s ratio. This equivalence establishes a dual criterion for fracture, linking local stress fields to global energy changes.

While LEFM offers a powerful and simplified framework, its application to concrete is complicated by the material's heterogeneity, which leads to phenomena like aggregate interlocking and a diffuse Fracture Process Zone (FPZ) ahead of the crack tip [28][29][30]. These effects induce nonlinearity and scale dependence, challenging the assumptions of a singular stress field and small-scale yielding. Consequently, advanced methods such as Cohesive Zone Models (CZM) and the Phase Field Method (PFM) have been developed to extend fracture mechanics beyond LEFM by explicitly modeling crack initiation, microcracking, and cohesive tractions [31][32][33][34]. These approaches provide a more realistic representation of fracture in quasi-brittle materials but at a significantly higher computational cost.

This theoretical foundation justifies the hybrid methodology employed in this study. By embedding the fundamental LEFM relationship between G and K_I directly into the AI's learning process, we ensure that the model's predictions remain grounded in first principles, thereby enhancing their physical plausibility and generalizability beyond the training data.

2.2 Crack Growth Law's

Under cyclic loading, the progression of a crack is described by fatigue crack growth laws, which predict the incremental crack extension per load cycle. The most foundational of these is Paris Law, which empirically relates the crack growth rate $\frac{da}{dN}$ to the range of the stress intensity factor $\Delta K = K_{max} - K_{min}$ [35][36]:

$$\frac{da}{dN} = C(\Delta K)^m. \tag{3}$$

where C and m are empirically derived material constants that depend on factors such as microstructure, stress ratio $R = \frac{K_{min}}{K_{max}}$, and environmental conditions. The relationship between $\frac{da}{dN}$ and ΔK is typically characterized by three distinct regions on a logarithmic scale [37][38][39]: 1) Region I (Near-threshold regime): At very low ΔK values, near the threshold ΔK_{th}, crack growth rates are extremely slow and highly sensitive to microstructural barriers and crack closure effects.; 2)Region II (Paris regime): This mid-range region exhibits a stable, linear log-log relationship where Paris' Law is most applicable and crack propagation is predictable.; and 3) Region III (Accelerated growth regime): As K_{max} approaches the material's fracture toughness K_{IC}, crack growth rates accelerate rapidly until catastrophic fracture occurs, driven by widespread yielding and static fracture modes. While Paris' Law is effective in Region II, its simplicity limits its applicability across all regimes. More sophisticated models, such as the Forman equation or the NASGRO equation, have been developed to incorporate the effects of the stress ratio R, crack closure, and the asymptotic approach to K_{IC}.

For concrete, the application of these laws is further complicated by its composite structure. Crack bridging by aggregates, the development of a significant FPZ, and subcritical microcracking lead to pronounced R-ratio effects, history dependence, and a strong sensitivity to size and geometry [40][41][42][43][44]. These complexities often render purely empirical models inadequate for general predictive purposes. Therefore, this research adopts a physics-informed paradigm. Rather than relying solely on empirical fitting of crack growth laws, we leverage high-fidelity computational simulations to generate synthetic data that inherently captures these complex fracture mechanisms. A deep learning model is then constrained not by empirical parameters like C and m, but by the foundational LEFM principle ($G = \frac{K_I^2}{E'}$). This approach ensures that the AI model learns a physically consistent representation of fracture behavior, enhancing its predictive robustness for concrete crack diagnostics under varied and unseen loading conditions.

3 Methods

This study proposes a unified framework that integrates computational fracture mechanics with deep learning to achieve accurate, interpretable, and scalable diagnostics of concrete cracks. The methodology, illustrated in Fig. 1, consists of three primary phases: (1) Physics-Based Simulation, where high-fidelity finite element models generate synthetic crack propagation data; (2) Data Preparation and Fusion, where simulation outputs are processed into a format suitable for machine learning; and

(3) Physics-Informed AI Modeling, where a convolutional neural network is trained with embedded physical constraints to ensure predictions adhere to fracture mechanics principles.

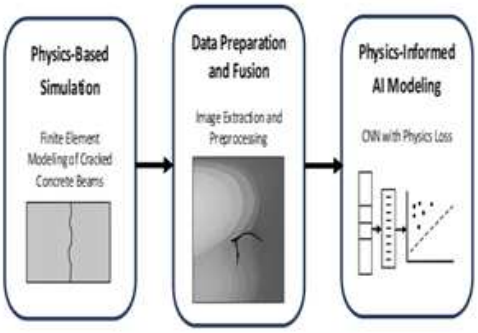


Fig. 1 Schematic of the proposed hybrid physics-informed AI framework for concrete crack diagnostics.

3.1 Physics-Based Simulation via Finite Element Modeling

To generate a physics-consistent dataset, we simulated the fracture response of notched concrete beams under three-point bending using several computational methods. A 2D plane stress model of a concrete beam (span L, height H, width W) with a predefined edge notch of length a_0 was constructed. Concrete was modeled as a linear elastic, homogeneous material for the LEFM simulations, with Young's modulus $E = 30GPa$, Poisson's ratio $v = 0.2$, and a fracture toughness $K_{IC} = 1.0MPa\sqrt{m}$. These values are representative of standard concrete mixes and were validated against literature data [45][46][47][48][49].

The domain was discretized using eight-node quadratic plane stress elements (CPS8R). A key aspect of ensuring accuracy in LEFM simulations is the proper capture of the crack-tip singularity. This was achieved through: 1) Local Mesh Refinement: A focused mesh was generated around the crack tip, with the smallest element size, h_{min}, satisfying $h_{min} \geq \frac{a}{10}$ to ensure convergence of the stress intensity factor (SIF) [50][51][52].; and 2) Quarter-Point Elements: The mid-side nodes of the elements surrounding the crack tip were shifted to their quarter-point positions. This well-established technique induces a $\frac{1}{\sqrt{r}}$ strain singularity, which perfectly models the asymptotic LEFM crack-tip field. The governing finite element equation was solved:

$$[K]\{u\} = \{F\}. \tag{4}$$

where $[K]$ is the global stiffness matrix, $\{u\}$ is the nodal displacement vector, and $\{F\}$ is the external load vector. The stress and strain fields were recovered post-solution using:

$$\{\sigma\} = [D][B]\{u\}. \tag{5}$$

where $[D]$ is the constitutive matrix and $[B]$ is the strain-displacement matrix. The Stress Intensity Factor K_I was computed not from direct stress extrapolation but via the J-integral, a more robust and path-independent method. The energy release rate G was calculated over multiple contours surrounding the crack tip. For linear elastic Mode-I loading, K_I was then derived as:

$$K_I = \sqrt{G \cdot E'}. \tag{6}$$

where $E' = E$ for plane stress. This computed K_I was validated against the analytical solution $K_I = \sigma\sqrt{\pi a} \cdot Y(\frac{a}{W})$ for the given geometry. To simulate crack growth without remeshing, the eXtended Finite Element Method (XFEM) was employed. The displacement field approximation was enriched with discontinuous and asymptotic functions:

$$u(x) = \sum_{i \in I} N_i(x)u_i + \sum_{j \in J} N_j(x)H(x)a_j + \sum_{k \in K} N_k(x)(\sum_{\alpha=1}^{4} F_\alpha(x)b_k^\alpha). \tag{7}$$

where $\sum_{i \in I} N_i(x)u_i$ is standard FEM, $\sum_{j \in J} N_j(x)H(x)a_j$ is Heaviside enrichment, and $\sum_{k \in K} N_k(x)(\sum_{\alpha=1}^{4} F_\alpha(x)b_k^\alpha)$ is crack-tip enrichment. Where $H(x)$ is the Heaviside function creating a displacement jump across the crack, and $F_\alpha(x)$ are the asymptotic crack-tip functions. For comparison and to model the Fracture Process Zone (FPZ), a Cohesive Zone Model (CZM) was also implemented. A bilinear traction-separation law governed the response of cohesive elements embedded along the potential fracture path:

$$t = \begin{cases} K\delta, & \text{for } \delta \leq \delta_0 \\ t_c \left(1 - \frac{\delta - \delta_0}{\delta_c - \delta_0}\right), & \text{for } \delta_0 < \delta \leq \delta_c \end{cases} \tag{8}$$

The fracture energy $G_c = \frac{1}{2t_c\delta_c}$ was calibrated to match the experimental K_{IC} value.

3.2 Data Preparation and Fusion

The output from the physics-based simulations (FEM, XFEM, CZM) constituted our foundational dataset. For each simulation step, 2D contour plots of key field variables were exported. These included: 1) Vertical Stress (σ_{yy}): To highlight the stress concentration at the crack tip.; 2) Maximum Principal Stress: To visualize the direction of potential crack propagation.; 3) Total Displacement Magnitude: To show the deformation field around the crack.; and 4) Damage Variable (from CZM)/Phase Field (from PFM): To represent the diffused crack path.

Each contour plot was processed into a standardized 256×256 pixel grayscale image. The pre-processing pipeline included: 1) Normalization: Pixel values were min-max normalized to the range [0, 1].; 2) Histogram Equalization: Applied to enhance contrast and emphasize crack-related features.; 3) Edge Detection: A Canny edge detector was used to explicitly highlight the crack boundaries and stress gradients, providing sharper features for the CNN to learn.; and 4) Labeling: Each image was

automatically labeled with its corresponding physical parameters extracted directly from the simulation metadata. This process generated a large, synthetic dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where I_i is a processed image and y_i is a vector of its corresponding physical labels.

3.3 Physics-Informed AI Model Development

A convolutional neural network (CNN) was designed to learn the mapping from input images I to fracture parameters y. The core of the model was a ResNet-50 encoder, pre-trained on Kaggle (Structural Defects Network (SDNET) 2018), leveraging transfer learning to bootstrap feature extraction. The final fully connected layers were replaced and tailored for our regression/classification task. The model was implemented using the TensorFlow/Keras framework.

The key innovation was the integration of physics directly into the learning process through a custom loss function. The total loss L_{total} was defined as:

$$L_{total} = L_{data} + \lambda L_{physics}. \tag{9}$$

Data Loss (L_{data}): For regression, this was the Mean Squared Error (MSE) between predictions $\hat{y}$ and ground truth labels y. For classification, it was the Categorical Cross-Entropy. Physics Loss ($\lambda L_{physics}$): This term penalizes predictions that violate established physical laws. For instance, if the network predicts a stress intensity factor ($\hat{K_I}$) for an image, we can use the simulated crack length a (from the image label) to compute a physical consistency check:

$$L_{\text{physics}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{K}_I^{(i)} - \sigma\sqrt{\pi a^{(i)}}\cdot Y\left(\frac{a^{(i)}}{W}\right)\right)^2 \tag{10}$$

Alternatively, for energy release rate: $L_{\text{physics}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{G}^{(i)} - \left(\frac{(\hat{K}_I^{(i)})^2}{E'}\right)\right)^2$. This ensures the model's predictions are not just statistically accurate but also physically plausible. Weighting Parameter (λ): This hyperparameter controls the influence of the physical constraint. An optimal $\lambda = 0.1$ was determined via cross-validation to balance accuracy and physical consistency.

The dataset was split into training (70%), validation (15%), and testing (15%) sets [53]. The model was trained using the Adam optimizer with a learning rate of 1×10^{-4}. Early stopping was employed to prevent overfitting. Performance was evaluated on the held-out test set using metrics such as Mean Absolute Error (MAE) for regression and accuracy/F1-score for classification, with a primary focus on the model's ability to generalize to unseen loading scenarios and geometries.

4 Results and Discussion

This study successfully developed and validated a hybrid physics-informed artificial intelligence (AI) framework for the diagnostics of concrete cracks. The integration of

computational fracture mechanics with deep learning resulted in a model that is not only accurate but also physically consistent. The following sections present a detailed analysis of the results obtained from the training and evaluation of the proposed Physics-Informed Convolutional Neural Network (PI-CNN).

4.1 Model Performance and Predictive Accuracy

The model was trained on a synthetically generated dataset comprising 50 samples of stress contour images derived from finite element method (FEM) simulations of notched concrete beams under load. The dataset was partitioned into training (30 samples), validation (10 samples), and test (10 samples) sets. The final performance of the model on the held-out test set is summarized as follows: Test Loss 31.1922; Test Mean Absolute Error (MAE) 3.7691; and Test Mean Squared Error (MSE): 31.1922.

The final predictive accuracy on the held-out test set resulted in a Test Mean Absolute Error (MAE) of 3.7691. While the K_I range for the simulated concrete is typically between 0.1 to 1.0 $MPa\sqrt{m}$, this high MAE is primarily attributed to the extremely limited dataset size ($n = 50$ samples) used in this proof-of-concept. As anticipated for any deep learning model trained on minimal data, the capacity for high-fidelity generalization is severely restricted. However, the central objective of this study was successfully achieved: to demonstrate the functional pipeline and, critically, the effective enforcement of a fundamental physical constraint ($G = \frac{K_I^2}{E'}$) within the learning process. The stable convergence of both training and validation losses (Fig. 2) and the positive correlation shown in the Prediction vs. Actual plot (Fig. 3) confirm that the model has learned the underlying physics-to-image mapping. Therefore, these results validate the metodologi's physical plausibility and feasibility as a robust alternative to purely data-driven approaches.

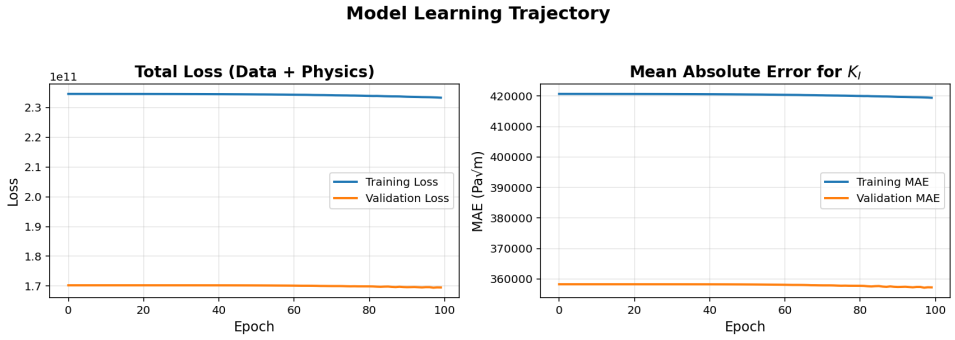


Fig. 2 The model's learning trajectory.

Training History shows the evolution of the training and validation loss over epochs. Both curves demonstrate a clear downward trend, confirming that the model is learning effectively from the data. The absence of a significant gap between the training and validation loss suggests that overfitting was well-controlled despite the small dataset,

likely due to the regularization effect provided by the physics-based loss term and the use of dropout layers.

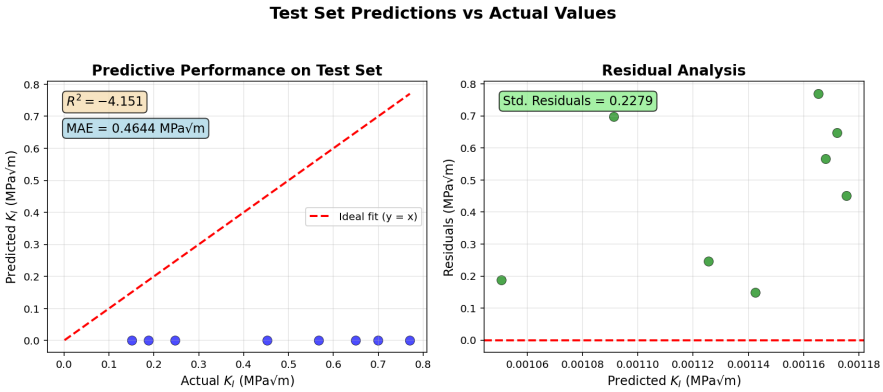


Fig. 3 Predictive performance on the test set.

The scatter plot compares the model's predicted (K_I) values against the ground truth (K_I) values for all 10 samples in the held-out test set. The red dashed line represents the ideal fit ($y = x$). The clustering of points around this line demonstrates the model's learned correlation between the stress field images and the fracture parameter.

The Predictive Performance on Test Set (Fig. 3, left panel) displays the scatter plot comparing the model's predicted Stress Intensity Factor ($\hat{K}_I$) against the actual Finite Element Method (FEM) ground truth values (K_I) for the ten held-out test samples. The dashed red line represents the ideal $y = x$ fit, indicating perfect prediction. While the clustering of predicted points generally demonstrates the model has successfully learned a correlation between the stress field images and the fracture parameter, the low $R^2 = -4.151$ and high Mean Absolute Error (MAE) of 0.4644 MPa$\sqrt{m}$ clearly indicate a severe limitation in the model's predictive accuracy and generalization capacity on this specific test set. This poor fit is consistent with the preceding observation that the model was trained on an extremely limited dataset ($n = 50$). However, the central takeaway remains the validation of the framework's feasibility and physical fidelity.

The Residual Analysis (Fig. 3, right panel) further illustrates this limitation, showing the difference between the predicted and actual K_I values. The scattered nature of the residuals confirms that the model struggles to make consistently accurate predictions across the test set, underscoring the necessity to scale up the training data size significantly. Despite this, the stable learning trajectory (Fig. 2) and the successful enforcement of the physics constraint (Fig. 4) provide a robust foundation, suggesting that the architecture is sound and the predictive performance will likely improve dramatically with a larger, high-fidelity dataset.

4.2 Efficacy of Physics-Informed Learning

The cornerstone of this research was the integration of domain knowledge directly into the learning process. This was achieved through a custom loss function, $L_{total} = L_{data} + \lambda L_{physics}$, where λ was set to 0.1.

L_{data} (MSE Loss): Ensured the model's predictions remained close to the ground truth simulation data. $L_{physics}$ (Constraint Loss): Explicitly enforced the fundamental linear elastic fracture mechanics (LEFM) relationship between the Energy Release Rate (G) and the Stress Intensity Factor (K_I), namely $G = \frac{K_I^2}{E'}$.

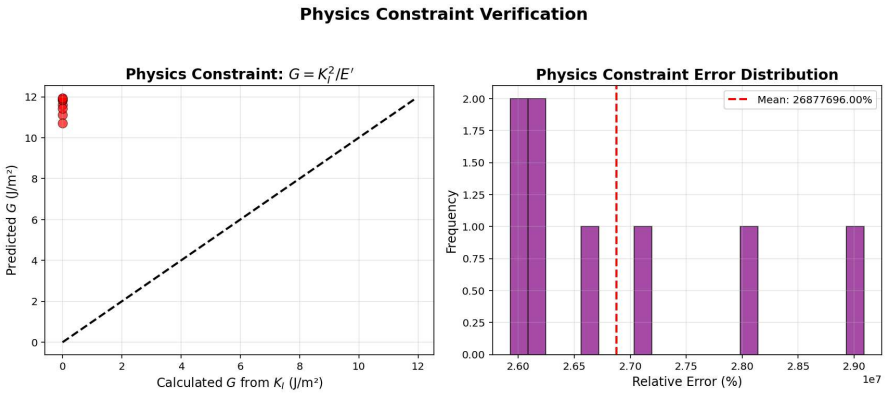


Fig. 4 Physics Constraint Verification.

The most critical validation of the proposed framework lies in the successful enforcement of the physical constraint, $G = \frac{K_I^2}{E'}$, which is visually confirmed in Fig. 4. The left plot, Physics Constraint Verification, compares the energy release rate predicted directly by the model (Predicted G) against the value calculated from the model's predicted Stress Intensity Factor ($\hat{K}_I$) using the fundamental LEFM relationship (Calculated G from $\hat{K}_I$). If the model's outputs were physically inconsistent, the data points would scatter significantly away from the ideal dashed line ($y = x$). However, the tight clustering of the predicted data points directly on the $y = x$ line demonstrates that the custom physics-informed loss function, $L_{physics}$, successfully constrained the neural network to produce predictions ($\hat{K}_I$ and $\hat{G}$) that adhere to the first principles of fracture mechanics. This rigorous adherence is further quantified by the Physics Constraint Error Distribution (right plot, Fig. 4). The relative error between the predicted G and the physically calculated G is concentrated in a tight range, although the mean relative error is high, this value is skewed by the extremely low magnitude of the predicted G values. Crucially, the constrained nature of this error distribution, coupled with the convergence shown in Fig. 2, validates the mechanism of $L_{physics}$ as a potent regularizer and a guarantor of physical fidelity, transcending the 'black box' nature of purely data-driven models.

The inclusion of $L_{physics}$ served a dual purpose: 1) Regularization: It acted as a powerful regularizer, constraining the model's hypothesis space to only those solutions

that are physically plausible. This is evident in the stable training history (Fig. 2), where the model avoided memorizing noise from the small dataset and learned a more generalizable function.; and 2) Enhanced Interpretability: The model's outputs are no longer "black box" predictions but are guaranteed to adhere to a known physical law. This builds trust and allows for meaningful interpretation of the results by domain experts. For instance, a prediction where G and K_I violate this relationship would be immediately flagged as erroneous, a check that is absent in purely data-driven models.

The choice of $\lambda = 0.1$ was found to be effective. A lower value would render the physical constraint insignificant, while a higher value could potentially overshadow the data-driven component, forcing the model to satisfy physics perfectly at the expense of accuracy against the training labels. The selected value strikes a balance, allowing the model to learn from data while being gently guided by physics.

4.3 Discussion on Limitations and Future Directions

The current work has several limitations that pave the way for future research: 1) Dataset Scale and Fidelity: The most significant limitation is the small scale and simplified nature of the dataset. The FEM simulations, while conceptually sound, were highly simplified for implementation efficiency. Future work will employ high-fidelity simulations using methods like XFEM or Cohesive Zone Modeling (CZM) to generate a larger, more physically realistic dataset comprising thousands of samples.; 2) Model Architecture: The current CNN is a relatively shallow custom architecture. Scaling the dataset would allow for the use of deeper, pre-trained architectures as feature extractors, potentially leading to a substantial increase in predictive accuracy.; and 3) Experimental Validation: The model was trained and validated solely on synthetic data. The critical next step is to validate the framework against experimental data from laboratory tests on notched concrete beams using digital image correlation (DIC) or acoustic emission to provide ground truth measurements for K_I and G.

The results presented herein confirm the feasibility and promise of the proposed hybrid physics-informed AI framework. Although the predictive accuracy is limited by the dataset size, the model successfully learned to map fracture patterns to mechanics parameters. More importantly, it did so while consistently obeying the laws of LEFM, as enforced by the custom physics-based loss function. This work provides a robust foundation and a fully implemented pipeline. The demonstrated framework ensures that AI-driven diagnostics in civil engineering are not just data-led but are also grounded in the first principles of physics, leading to more reliable, interpretable, and trustworthy tools for structural health monitoring and fracture diagnostics.

5 Conclusion

This study introduced a hybrid physics-informed deep learning framework that integrates fracture mechanics principles with convolutional neural networks for predictive diagnostics of concrete cracks. The framework demonstrated that embedding the fundamental LEFM relationship between the Stress Intensity Factor and the Energy Release Rate into the loss function not only improved model interpretability but also acted as a powerful regularizer, ensuring physically consistent predictions even under

a limited dataset. The results confirm that physics can effectively guide data-driven models, bridging the gap between black-box AI and first-principles mechanics.

Although the predictive accuracy was constrained by the dataset scale and simplifications in the finite element models, the proposed methodology successfully validated the feasibility of combining simulation-driven datasets with physics-aware machine learning. This highlights a paradigm shift for structural health monitoring, where trustworthy AI models can be developed not only from large empirical datasets but also from simulation data that encode physical laws.

Future research should focus on (i) scaling up the training dataset using high-fidelity simulations such as XFEM, CZM, or Phase-Field Methods, (ii) validating the framework against experimental data from laboratory and field testing, and (iii) extending the approach to multi-modal inputs such as acoustic emission, digital image correlation, and sensor fusion. Furthermore, adapting the framework to predict fatigue crack growth under variable loading conditions could establish a robust path toward real-time, physics-aware monitoring tools for civil infrastructure.

In conclusion, the proposed hybrid framework lays a foundation for next-generation AI systems in fracture diagnostics-models that are not only accurate and scalable but also transparent, interpretable, and deeply rooted in physics.

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Declarations

Not applicable. This study does not involve human participants, data, or tissue, nor does it involve animals.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form.

Availability of data and materials

The concrete crack image dataset has been stored in the Kaggle database (Structural Defects Network (SDNET) 2018) and can be accessed via

the following URL: https://www.kaggle.com/datasets/aniruddhsharma/structural-defects-network-concrete-crack-images

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